

tf.compat.v1.constant Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/constant

Creates a constant tensor.

Function Signatures

```
tf.compat.v1.constant( value, dtype=None, shape=None,  
name='Const', verify_shape=False ) -> Union[tf.Operation,  
ops._EagerTensorBase]
```

Code Examples

```
tf.compat.v1.constant(  
    value, dtype=None, shape=None, name='Const', verify_shape=False  
) -> Union[tf.Operation, ops._EagerTensorBase]
```

```
tf.compat.v1.constant(  
    value, dtype=None, shape=None, name='Const', verify_shape=False  
) -> Union[tf.Operation, ops._EagerTensorBase]
```

```
tf.Operation
```

```
# Constant 1-D Tensor populated with value list.  
tensor = tf.constant([1, 2, 3, 4, 5, 6, 7]) => [1 2 3 4 5 6 7]  
# Constant 2-D tensor populated with scalar value -1.  
tensor = tf.constant(-1.0, shape=[2, 3]) => [[-1. -1. -1.]  
[-1. -1. -1.]]
```

```
# Constant 1-D Tensor populated with value list.  
tensor = tf.constant([1, 2, 3, 4, 5, 6, 7]) => [1 2 3 4 5 6 7]  
# Constant 2-D tensor populated with scalar value -1.  
tensor = tf.constant(-1.0, shape=[2, 3]) => [[-1. -1. -1.]  
[-1. -1. -1.]]
```

Documentation

Creates a constant tensor.

The resulting tensor is populated with values of type `dtype`, as specified by arguments `value` and (optionally) `shape` (see examples below).

The argument value can be a constant value, or a list of values of type dtype. If value is a list, then the length of the list must be less than or equal to the number of elements implied by the shape argument (if specified). In the case where the list length is less than the number of elements specified by shape, the last element in the list will be used to fill the remaining entries.

The argument shape is optional. If present, it specifies the dimensions of the resulting tensor. If not present, the shape of value is used.

If the argument dtype is not specified, then the type is inferred from the type of value.

For example:

tf.constant differs from tf.fill in a few ways:

- tf.constant supports arbitrary constants, not just uniform scalar

Tensors like tf.fill.

- tf.constant creates a Const node in the computation graph with the

exact value at graph construction time. On the other hand, tf.fill creates an Op in the graph that is expanded at runtime.

- Because tf.constant only embeds constant values in the graph, it does

not support dynamic shapes based on other runtime Tensors, whereas tf.fill does.

Returns

Raises

Generated on 10/25/2025 | TensorFlow Documentation Scraper